

SDA GIS Full Manual Review Methodology

Written by: Payman Alemi

Last Updated: 5/30/2025

Table of Contents

1	ArcGIS Pro Portion.....	1
1.1	Initial Setup	1
1.2	Convert the eWRIMS flat file to a shapefile.....	5
1.3	Add the PLSS information from the BLM PLSS shapefile to the eWRIMS PODs	7
1.4	Filter to just the PODs in your watershed.....	10
1.4.1	Create a shapefile that just intersects the watershed	10
1.4.2	Relate the RR_Section_Intersect_BLM_2024 shapefile to the ewrims_statewide_pods_PLSS_spatial_join shapefile	13
1.5	Determine Edge Cases for Manual Review	17
1.5.1	Create the watershed boundary layer	17
1.5.2	Create the watershed interior buffer, interior, and exterior buffer layers	18
1.5.3	Eliminating Unnecessary PODs from the Manual Review	20
2	Manual Review Process using ArcGIS and Excel	22
2.1	Import the Manual Review POD List into Excel	22
2.2	Bulk Download all eWRIMS documents for your watershed	23
2.3	How to fill out the manual review spreadsheet	23
3	Appendix	28
3.1	Using Power Query and M Scripts	28
3.2	Using USGS StreamStats Tool	31
3.3	Using EarthPoint to convert Northings/Easting into Latitude/Longitude.....	34
3.4	Finding POD Coordinates in the absence of latitude/longitude and northing/easting	36

1 ArcGIS Pro Portion

1.1 Initial Setup

- 1) Create a local geodatabase just for this project, so that you have local storage space for you GIS files—this will substantially expedite processing in ArcGIS Pro. Normally, if you create a project in ArcGIS Pro, then ArcGIS Pro will automatically create an associated local geodatabase, which you can see in the *Contents* pane in Figure 1.

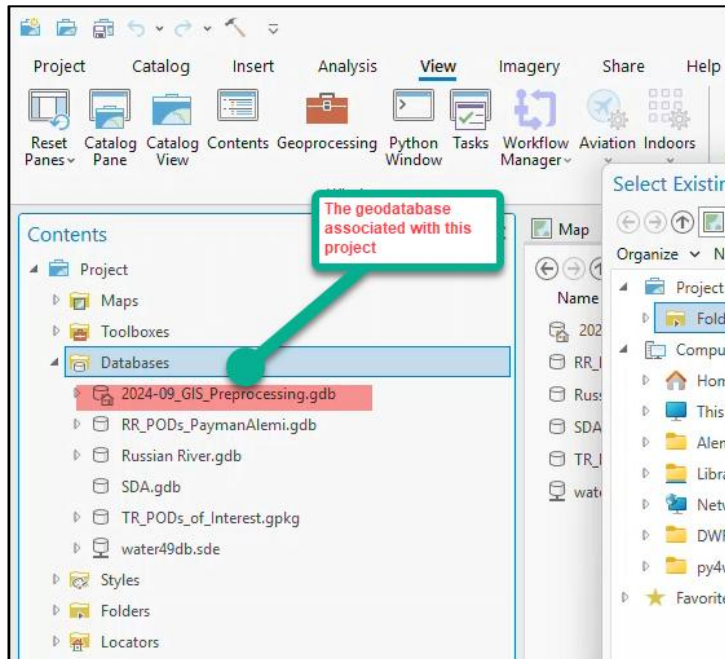


Figure 1. The geodatabase tied to the current project map.

- 2) Add the Flat_File_eWRIMS-YYYY-MM-DD.csv (“eWRIMS flat file”) generated by the GIS_POD_Flat_File_Prep.R script to your project map and to your local geodatabase.
- 3) Using File Explorer, navigate to the SDA_Public_gdb.zip file in the DWRAT_DataScraping/Demand/InputData folder. Right-click to unzip the file and save the entire geodatabase to your computer as SDA_public.gdb—keep it off SharePoint and/or other cloud services to avoid syncing issues.
- 4) In ArcGIS Pro, navigate to the View pane and select Catalog Pane and scroll to Databases:

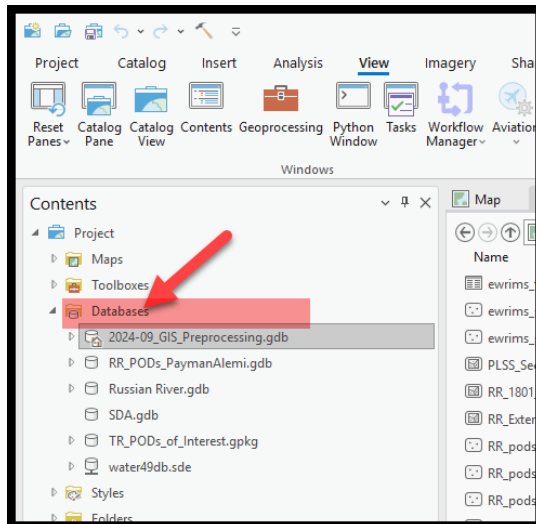


Figure 2. Database button in Catalog Pane.

- 5) Right-click on *Databases* and select *Add Database* from the pop-up menu:

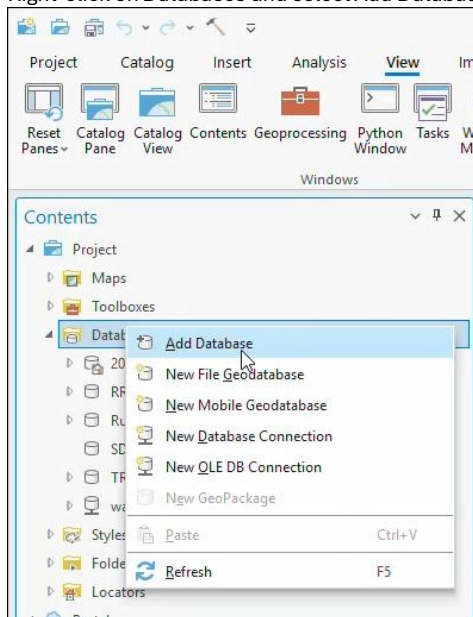


Figure 3. Adding an existing database in ArcGIS Pro.

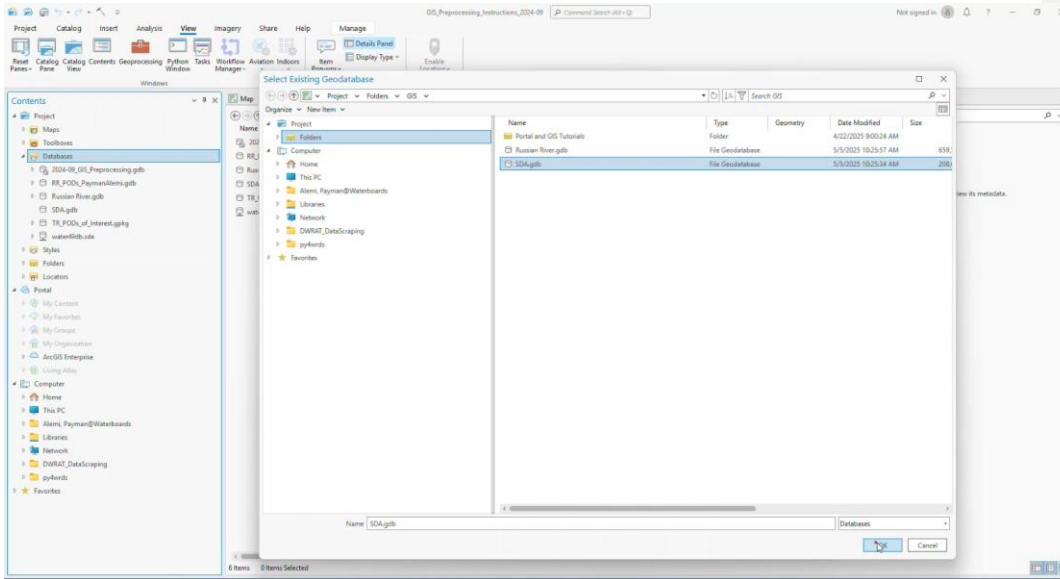


Figure 4. Select SDA_public.gdb from your saved location.

- 6) Then navigate to *SDA_public.gdb* in the pop-up window, click on it, and click **OK** to load it into your list of databases:
- 7) In the ArcGIS Pro ribbon, go to **View** → **Catalog** view and then navigate to the **Contents** pane. You should see *Databases*—right-click on it to reveal *SDA_public.gdb*.
- 8) Click on *SDA_public.gdb* and ArcGIS Pro will reveal all the shapefiles contained in the geodatabase like so:

Commented [PA1]: @Aleml, Payman@Waterboards
Replace the screenshot when you create SDA_public.gdb

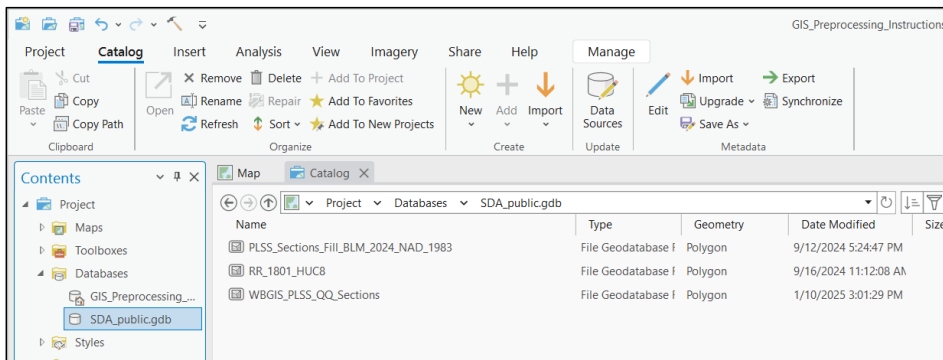


Figure 5. Shapefiles within SDA_public.gdb

- 9) Right-click on the BLM PLSS Quarter-Quarter sections shapefile (*WBGIS.PLSS_QQ_Sections*) from *SDA_public.gdb* and select **Add to Map** from the pop-up menu to add this shapefile to your project map and your local geodatabase:

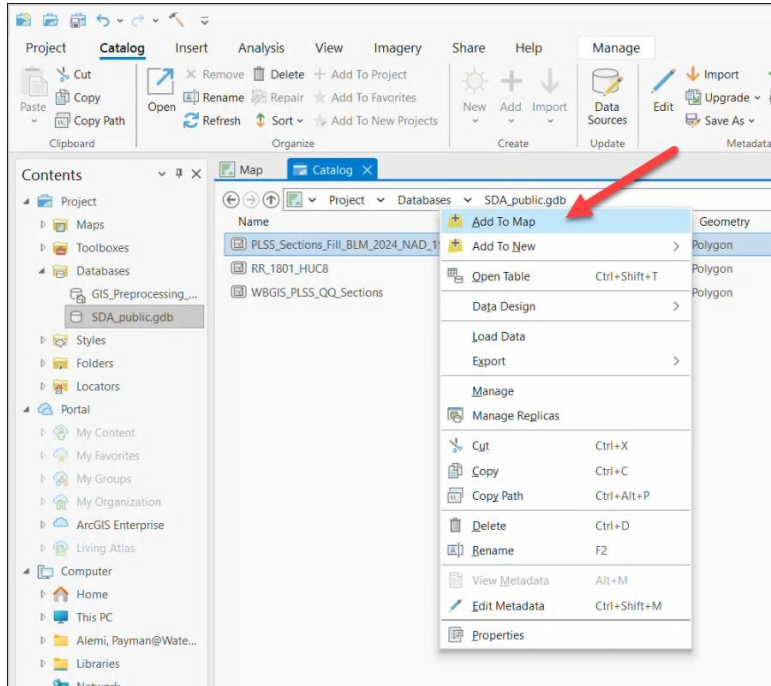


Figure 6. Adding a shapefile from a geodatabase.

- 10) Add the *PLSS_Sections_Fill_BLM_2024_NAD_1983* shapefile to your project map too—this shapefile contains the public land survey sections, townships, and ranges for the state of California.
- 11) Add the shapefile corresponding to your watershed to the project map and local geodatabase. If you would like to use the Russian River watershed as an example, add the *RR_1801_HUC8* shapefile from *SDA_public.gdb* to your project map.

1.2 Convert the eWRIMS flat file to a shapefile

- 1) Run the *Export Table* Geoprocessing tool. This tool allows you to change the data types for the fields in the eWRIMS flat file. At the outset, the default data type for all fields is text, which will prevent us from applying the later steps.
 - a) Find the tool by navigating to the **Analysis Pane**, clicking on the **Tools** icon, and then searching for *Export Table*. Select *Export Table (Conversion Tools)* from the resulting menu. I recommend adding the tool as a project favorite (right click and select “Add to my Favorites”), so you don’t have to search for it every time:

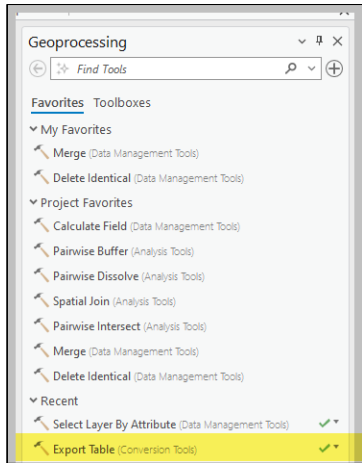


Figure 7. Export Table Tool

- Enter these parameters into the *Export Table* tool:
Input Table: Flat_File_eWRIMS_YYYY-MM-DD.csv
Output Table: ewrims_flat_file_table (save this table to your local geodatabase).
- Click on the *Fields* button to expand the *Field Map*. Then click on individual fields as shown below to change the *type*.

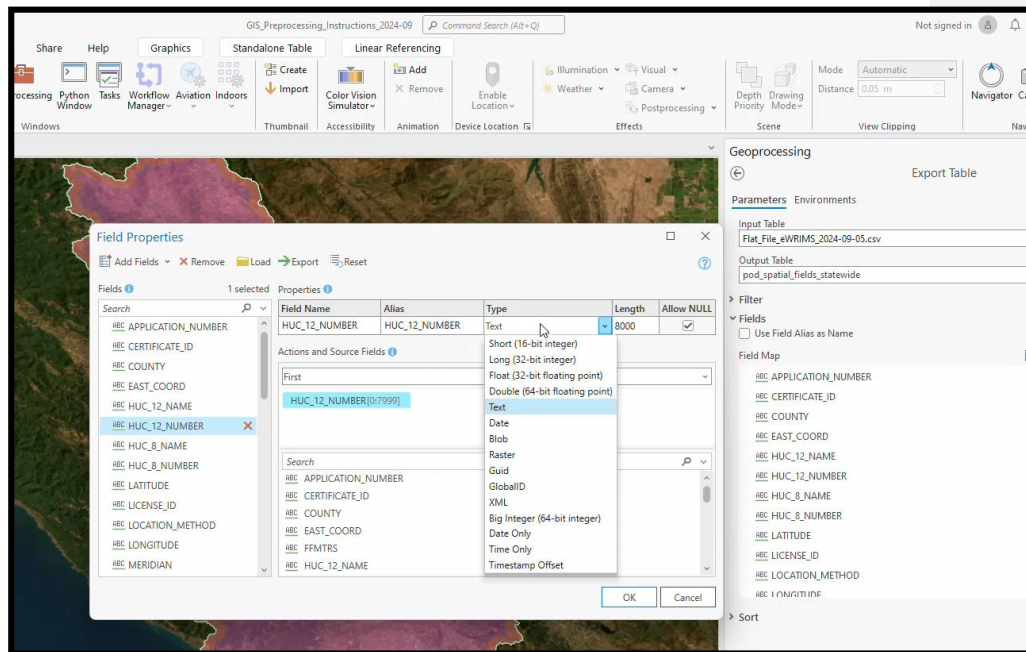


Figure 8. Adding Fields to an Attribute Table

7 of 38

- d) Change these fields as follows:
 - HUC12_NUMBER:** change type to *Long (32-bit integer)*
 - HUC8_NUMBER:** change type to *Long (32-bit integer)*
 - LATITUDE:** change type to *Double (64-bit floating point)*
 - LONGITUDE:** change type to *Double (64-bit floating point)*
 - RANGE_NUMBER:** change type to *Short (16-bit integer)*
 - SECTION_NUMBER:** change type to *Short (16-bit integer)*
 - TOWNSHIP_NUMBER:** change type to *Short (16-bit integer)*
 - e) Click *Run* to generate *ewrims_flat_file_table*.
- 2) Run the *XY Table to Point* geoprocessing tool—this tool actually converts *ewrims_flat_file_table* into a shapefile.
- a) In the ribbon, navigate to *Analysis* → *Tools* → *XY Table to Point (Data Management Tools)*
 - b) Enter the following parameters and click *Run*. Figure 9 shows a screenshot for clarity, but the written instructions take precedence because it was impractical to retake a screenshot every time the instructions were slightly tweaked.

Input Table: *ewrims_flat_file_table*

Output Feature Class: *ewrims_statewide_pods*

X Field: *LONGITUDE*

Y Field: *LATITUDE*

Z Field: Leave blank

Coordinate System: *GCS_North_American_1983*—which you can click on the globe and navigate to select; it may appear with the alias *NAD 1983*.

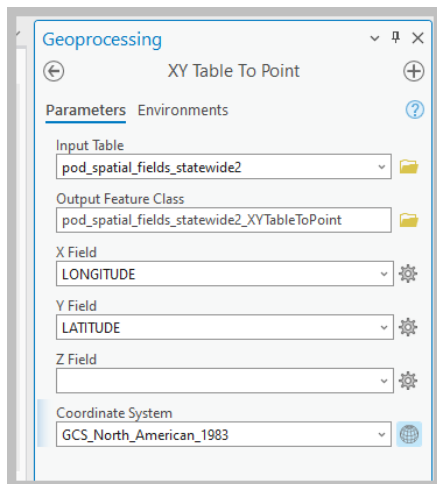


Figure 9. XY Table to Point Tool

1.3 Add the PLSS information from the BLM PLSS shapefile to the eWRIMS PODs

- 1) Spatially join the PLSS BLM shapefile to the *ewrims_statewide_pod* shapefile

7 of 38

8 of 38

- In the ribbon, navigate to *Analysis* → *Tools* → *Spatial Join (Analysis Tools)*
- Enter these parameters and click *Run*. The screenshot shows the tool's appearance, but the written instructions below take precedent:

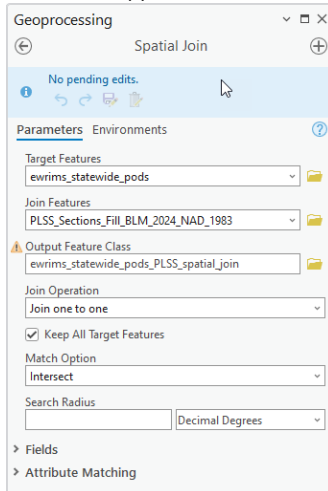


Figure 10. Spatial Join Tool

Target Features: ewrims_statewide_pods

Join Features: PLSS_Sections_Fill_BLM_2024_NAD_1983

Output Feature Class: ewrims_statewide_pods_PLSS_spatial_join

Join Operation: One to Many

Checkmark: Keep All Target Features

Match Option: Intersect

Search Radius: Leave the quantity field blank but select **Decimal Degrees** for the units.

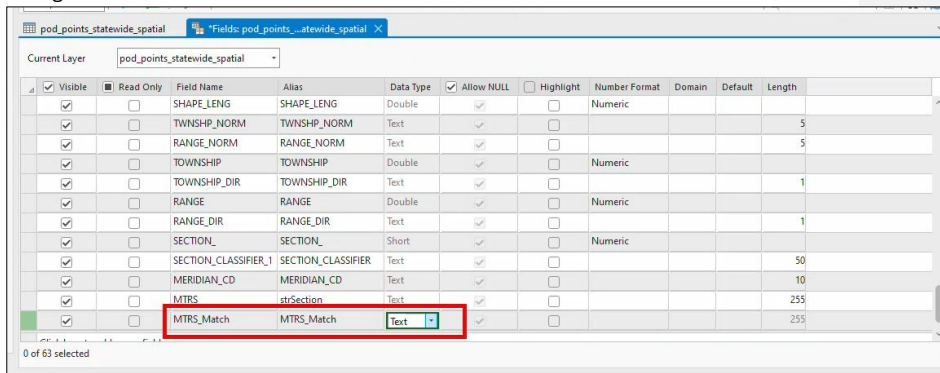
- Right-click on the *ewrims_statewide_pods_PLSS_spatial_join* shapefile and open the attribute table. Check that both the *MTRS* and *FFMTRS* fields are present.
- Add the *MTRS_Match* field to the attribute table by clicking the *Add Field* button.

OBJECTID	Shape	Join_Count	TARGET_FID	APPLICATION_NUMBER	CERTIFICATE_ID	COUNTY	EAST_COORD	HUC_12_NAME	HUC_12_NUMBER	HUC_8_NAME
1	Point	1	1	A000016	000041	Los Angeles	6474612.147768	Headwaters Amargosa...	18090206	Antelope-Fremont Vi
2	Point	1	2	A000016	000041	Los Angeles	6474612.147768	Headwaters Amargosa...	18090206	Antelope-Fremont Vi
3	Point	1	3	A000016	000041	Los Angeles	6474614.806227	Headwaters Amargosa...	18090206	Antelope-Fremont Vi
4	Point	1	4	A000016	000041	Los Angeles	6474612.147768	Headwaters Amargosa...	18090206	Antelope-Fremont Vi
5	Point	1	5	A000018	002871	Glenn	6547331	Murphy Slough-Sacra...	18020157	Big Chico Creek-Sacr
6	Point	0	6	NA	NA	NA	0	NA	0	NA
7	Point	0	7	NA	NA	NA	0	NA	0	NA
8	Point	1	8	A000023	001986	Madera	6793175	Lost Lake-San Joaquin...	18040001	Middle San Joaquin
9	Point	1	9	A000026	000036	Lake	6346157.876052	Kelsey Creek	18020116	Upper Cache
10	Point	1	10	A000027	003165	Colusa	6574809.313105	Packer Lake-Sacrament...	18020104	Sacramento-Stone C
11	Point	1	11	A000027	003165	Colusa	6564109.470004	Packer Lake-Sacrament...	18020104	Sacramento-Stone C
12	Point	1	12	A000027	003165	Glenn	6559684.850572	Packer Lake-Sacrament...	18020104	Sacramento-Stone C
13	Point	1	13	A000042	000211	Mono	7092376.316981	Lower Rock Creek	18090102	Crowley Lake

Figure 11. Add MTRS_Match Field

8 of 38

- 4) Then enter these parameters for the *Field Name*, *Alias*, and *Data Type* and save the changes to the attribute table.



Visible	Read Only	Field Name	Alias	Data Type	Allow NULL	Highlight	Number Format	Domain	Default	Length
☒	☐	SHAPE_LENGTH	SHAPE_LENGTH	Double	☒	☐	Numeric			
☒	☐	TOWNSHIP_NORM	TOWNSHIP_NORM	Text	☒	☐				5
☒	☐	RANGE_NORM	RANGE_NORM	Text	☒	☐				5
☒	☐	TOWNSHIP	TOWNSHIP	Double	☒	☐	Numeric			
☒	☐	TOWNSHIP_DIR	TOWNSHIP_DIR	Text	☒	☐				1
☒	☐	RANGE	RANGE	Double	☒	☐	Numeric			
☒	☐	RANGE_DIR	RANGE_DIR	Text	☒	☐				1
☒	☐	SECTION_	SECTION_	Short	☒	☐	Numeric			
☒	☐	SECTION_CLASSIFIER_1	SECTION_CLASSIFIER	Text	☒	☐				50
☒	☐	MERIDIAN_CD	MERIDIAN_CD	Text	☒	☐				10
☒	☐	MTRS	strSection	Text	☒	☐				255
☒	☐	MTRS_Match	MTRS_Match	Text	☒	☐				255

Figure 12. Field Properties of MTRS_Match

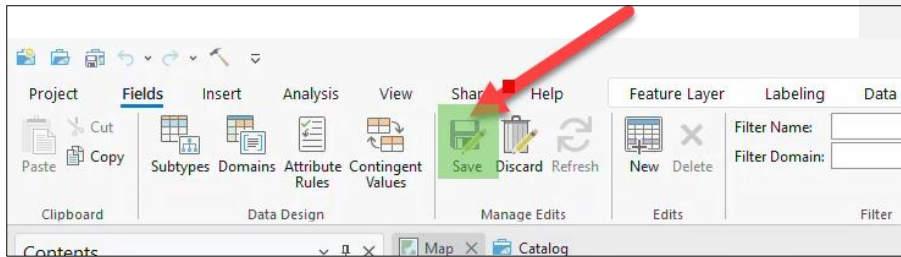


Figure 13. Saving changes to an attribute table

- 5) Back in the attribute table, right-click on the *MTRS_Match* field, and click on *Calculate Field*:

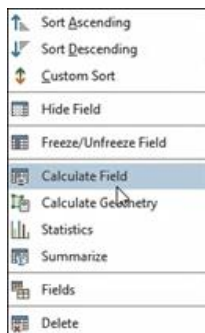


Figure 14. Calculate Field for MTRS_Match

- 6) Calculate **MTRS_Match** by using the Python expression below and set the data type to text; otherwise, the field calculator will fail.

10 of 38

Select *Python* as the **Expression Type**. Toggle on the *Enable Undo* button in case the code fails and click *OK*.

```
MTRS_Match = reclass(!FFMTRS!,!MTRS!)
```

Code Block

```
def reclass(FFMTRS, MTRS):  
    if FFMTRS == MTRS:  
        return "Y"  
    else:  
        return "N"
```

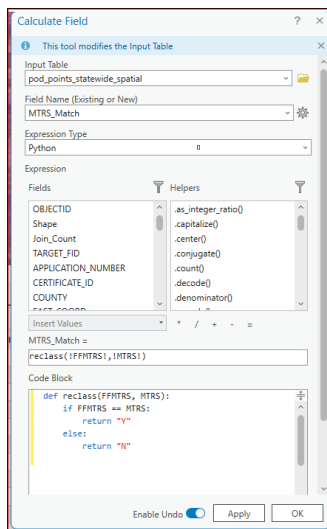


Figure 15. Python code block for MTRS_Match

1.4 Filter to just the PODs in your watershed

You can apply the steps in this section to any watershed. Just use the shapefiles that correspond to your watershed instead, but the sequence is the same. The example here uses the Russian River watershed.

1.4.1 Create a shapefile that just intersects the watershed

- 1) Select *PLSS_Section_Fill_BLM_2024_NAD_1983* by location. In the ribbon, navigate to *Map* → *Select by Location*
- 2) Enter these parameters into the *Select by Location* wizard and click *OK*.
Input Features: *PLSS_Sections_Fill_BLM_2024_NAD_1983*
Relationship: *Intersect*
Selecting Features: *RR_1801_HUC8* (or the polygon shapefile that represents your watershed)
Search Distance: Leave the quantity blank and enter **Decimal Degrees** for the units
Selection: *New Selection*

10 of 38

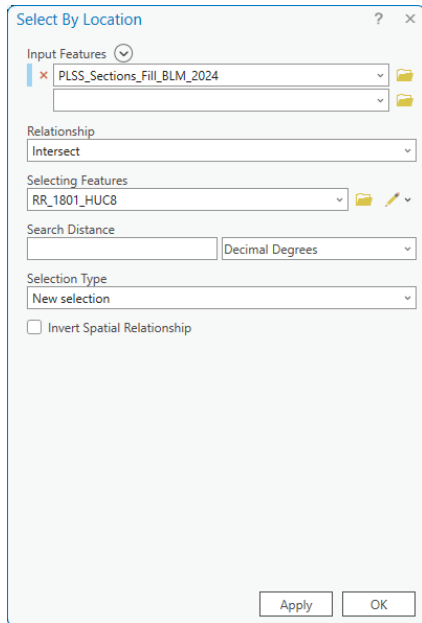


Figure 16. Select by Location to your watershed

- 3) The portion of the *PLSS_Sections_Fill_BLM_NAD_1983* shapefile that overlaps with the *RR_1801_HUC8* shapefile should be selected. Right-click on the *PLSS_Sections_Fill_BLM_NAD_1983* shapefile in the *Contents* pane and click on *Selection* → *Make Layer from Selected Features*.
- 4) Rename the newly created layer *XX_Section_Intersect_BLM_2024* with XX representing your watershed alias. For the Russian River, we are using RR—hence, we are renaming the file *RR_Section_Intersect_BLM_2024*.
- 5) Save *XX_Section_Intersect_BLM_2024* to your local geodatabase. Right-click on the shapefile and go to *Data* → *Export Features* and a dialog box will appear.
- 6) Input these parameters into the *Export Features* dialog box
 - a. *Input Features*: *XX_Section_Intersect_BLM_2024*
 - b. *Output Feature Class*: *XX_Section_Intersect_BLM_2024*

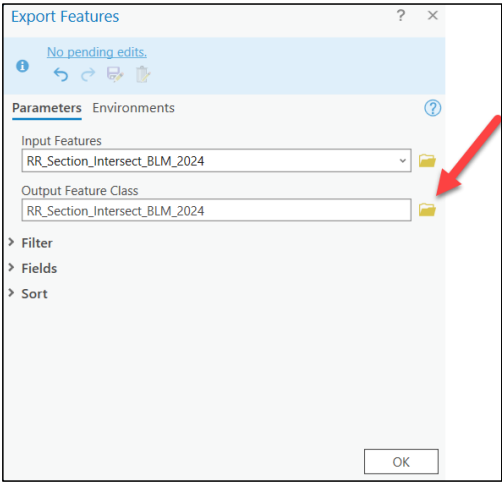


Figure 17. Export Features Dialog Box

The output feature class should be saved by default to your local geodatabase but to verify, you can click on the folder icon next to the field, navigate to the geodatabase, and select it from the search window that emerges:

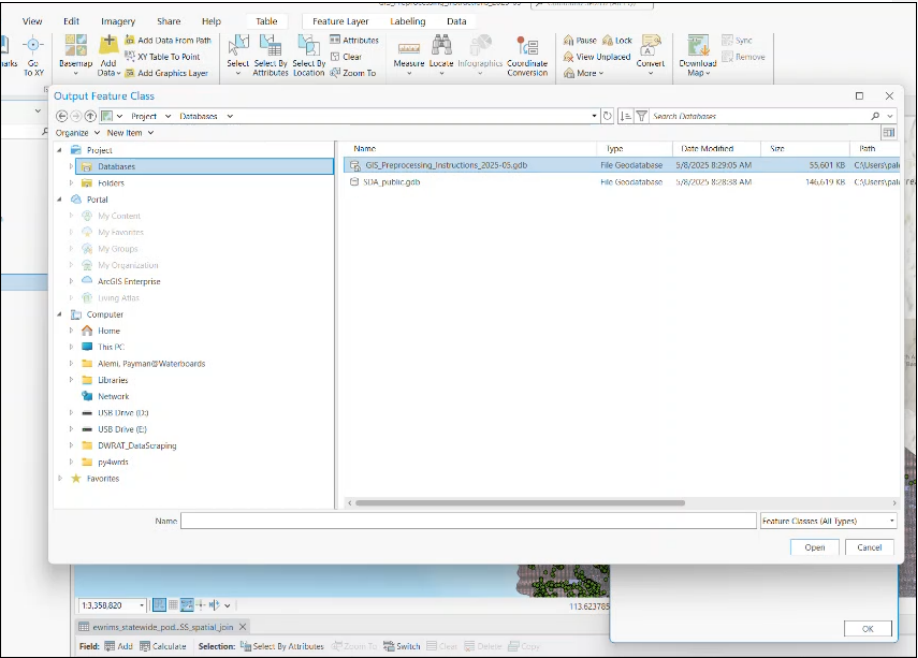


Figure 18. Export Features Search Window

13 of 38

1.4.2 Relate the XX_Section_Intersect_BLM_2024 shapefile to the ewrims_statewide_pods_PLSS_spatial_join shapefile

- 1) Right-click **XX_Section_Intersect_BLM_2024** and navigate to *Joins and Relates* → *Add Relate*
- 2) Input these parameters into the *Add Relate* wizard and click *OK*:
Layer Name: or Table View: RR_Section_Intersect_BLM_2024
Input Relate Field: MTRS
Relate Table: ewrims_statewide_pods_PLSS_spatial_join
Output Relate Field: MTRS
Relate Name: MTRS_Relate
Cardinality: One to many

The screenshot shows the 'Add Relate' dialog box with the following settings:

- Layer Name or Table View:** RR_Section_Intersect_BLM_2024
- Input Relate Field:** MTRS
- Relate Table:** ewrims_statewide_pods_PLSS_spatial_join
- Output Relate Field:** MTRS
- Relate Name:** MTRS_Relate
- Cardinality:** One to many

Figure 19. Adding MTRS_Relate

- 3) Right-click on the *RR_Section_Intersect_BLM_2024* layer in the *Contents* pane and go to *Properties* → *Selection*
 - A. Check the box to *automatically select related data*
 - B. Click *OK*.

13 of 38

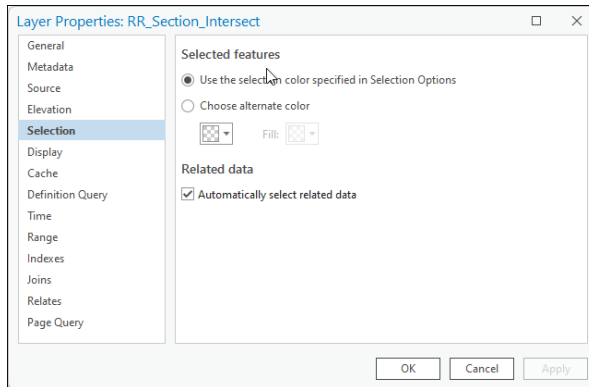
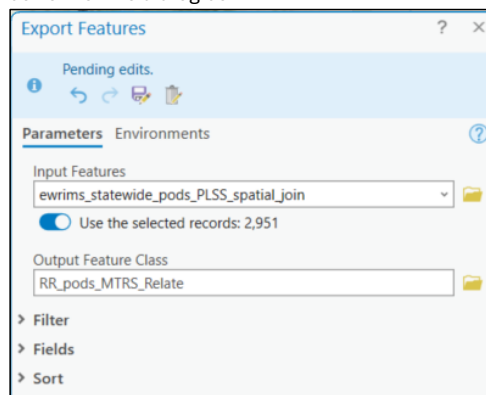


Figure 20. Automatically selecting related data

- 4) **Create *XX_pods_MTRS_Relate* shapefile** (remember XX is your watershed alias; for the Russian River, it's RR)
 - A. Right-click the *XX_Section_Intersect_BLM_2024* layer in the *Contents* pane and click on the *Attribute Table*.
 - B. Select all the polygons in the *XX_Section_Intersect_BLM_2024* layer by clicking anywhere on the attribute table and clicking **Ctrl+A** on your keyboard.
 - C. Since we checked the *Automatically select related data* checkbox earlier, all the related PODs in the *ewrims_statewide_pods_PLSS_spatial_join* layer will also be selected.
 - D. Right-click the *ewrims_statewide_pods_PLSS_spatial_join* layer in the *Contents* pane and go to *Data* → *Export Features*
 - E. Toggle on the *Use the selected records* button and export the feature class as *XX_pods_MTRS_Relate* to your local geodatabase. Click **OK** at the bottom of the dialog box.

Figure 21. Creating the *RR_pods_MTRS_Relate* layer

- F. Clear all selections from your ArcGIS Pro project.

- 5) **Create the *XX_pods_FFMTS_Relate* shapefile**

15 of 38

Select PODs where the *FFMTRS* field in *ewrims_statewide_pods_PLSS_spatial_join* matches the *MTRS* field in the *XX_Section_Intersect_BLM_2024* shapefile.

- A. Right-click on the *ewrims_statewide_pods_PLSS_spatial_join* shapefile and go to *Joins and Relates* → *Remove All Relates*.
- B. Right-click on the *XX_Section_Intersect_BLM_2024* shapefile and remove all the relates.
- C. Right-click on the *XX_Section_Intersect_BLM_2024* shapefile and go to *Properties* → *Selection*. Underneath the *Related Data* section, ensure that the *Automatically select related data* checkbox is still checked.
- D. Right-click on the *XX_Section_Intersect_BLM_2024* again and go to *Joins and Relates* → *Add Relate* and enter these parameters, and then click *OK*.

Layer Name or Table View: *XX_Section_Intersect_BLM_2024*

Input Relate Field: *MTRS*

Relate Table: *ewrims_statewide_pods_PLSS_spatial_join*

Output Relate Field: *FFMTRS*

Relate Name: *FFMTRS_Relate*

Cardinality: *One to Many*

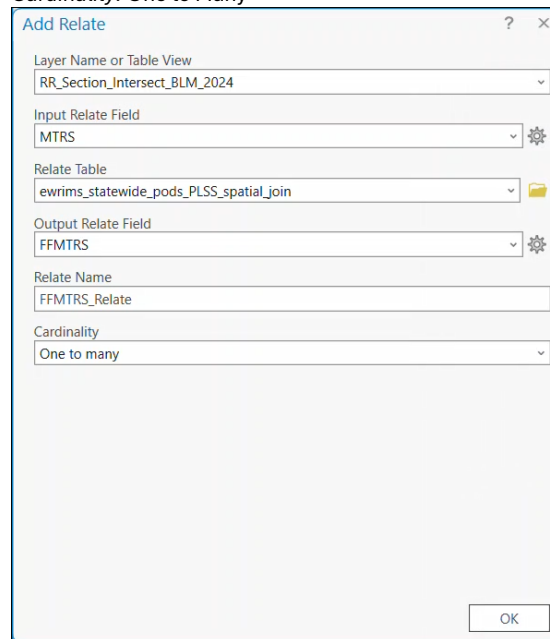


Figure 22. Creating *FFMTRS_Relate*

- E. Open the attribute table of the *XX_Section_Intersect_BLM_2024* shapefile and select all the polygons.
- F. Right-click on the *ewrims_statewide_pods_PLSS_spatial_join* shapefile and go to *Data* → *Export Features*
 - i. Toggle on the *use selected records* button and export as *XX_pods_FFMTRS_Relate* to your local geodatabase.

15 of 38

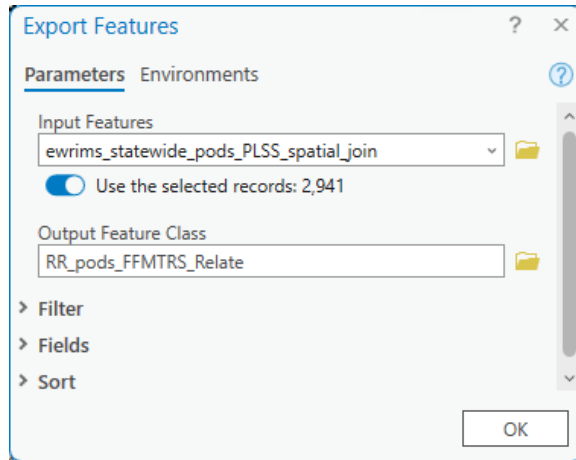


Figure 23. Creating the RR_pods_FFMTRS_Relate layer

G. Clear all the selections from your ArcGIS Pro project.

- 6) Merge the XX_pods_MTRS_Relate and XX_pods_FFMTRS_Relate shapefiles.
 A. Go to the *Analysis Pane* and navigate to the *Merge (Data Management Tools)* tool and enter these parameters:
Input datasets: XX_pods_FFMTRS_relate, XX_pods_MTRS_relate
Output dataset: XX_pods_merged

Commented [PA2]: @Alemi, Payman@Waterboards start here in the afternoon.

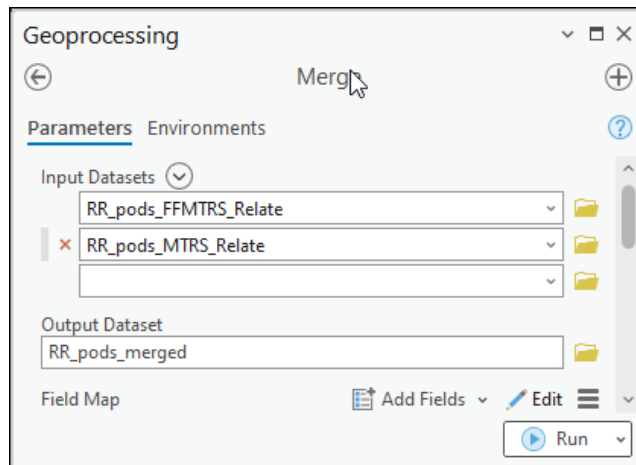


Figure 24. Merge (Data Management Tools) Tool

The total records should be the sum of the records in the attribute tables of the two layers.

- 7) Delete identical records from the XX_pods_merged shapefile with the *Delete Identical (Data Management Tools)* tool based on the *POD_ID* field and click *Run*.

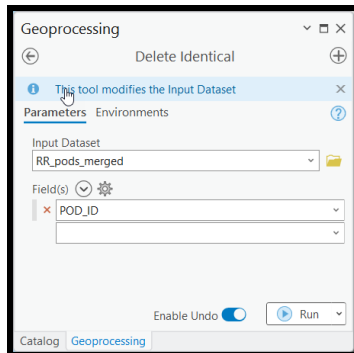


Figure 25. Delete Identical (Data Management Tools) Tool

- 8) You have now captured all PODs that geospatially may be located in your watershed. As of 5/29/2025, this series of steps yield 2,919 PODs in the *RR_pods_merged* shapefile. The number of PODs will vary for other watersheds; if you have thousands of PODs, a full manual review will be infeasible, so SDA recommends whittling down the list to edge cases as indicated in the next section.
- 9) Add a new text field, *Keep_POD*, to the attribute table of the *XX_pods_merged* shapefile.
- 10) Export the attribute table of the *XX_pods_merged* shapefile as a csv and add a datestamp as the suffix, e.g., *XX_pods_merged_YYYY-MM-DD.csv*, into the *DWRAT_DataScraping/Demand/OutputData* folder. This CSV will serve as the full list of possible PODs for your watershed and after you finish your manual review, you will remove the out of watershed PODs from this CSV to create your final watershed POD List.

1.5 Determine Edge Cases for Manual Review

1.5.1 Create the watershed boundary layer

- 1) Load the *RR_1801_HUC8* shapefile from the SDA GIS geodatabase. If you are creating a boundary polyline for another watershed, then load the corresponding HUC8 or HUC12 shapefile. The remaining steps are identical for other watersheds.
- 2) Navigate to the *Analysis Pane* → *Tools* and search for *Polygon to Line*. This tool is only licensed for use in **ArcGIS Pro – Advanced**.
- 3) Enter these parameters and click *Run*:
 - Input Features*: *RR_1801_HUC8* (or your watershed polygon)
 - Output Feature Class*: *XX_WB* (save to your local geodatabase)
 - Checkmark Identify and store polygon neighboring information*.

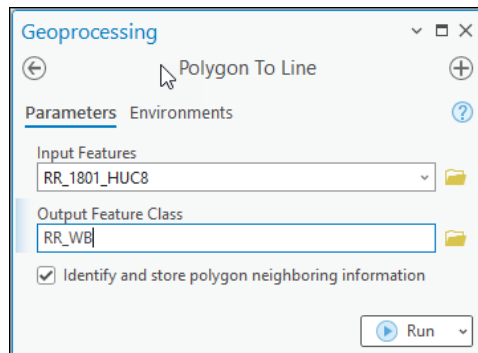


Figure 26. Polygon To Line Tool

1.5.2 Create the watershed interior buffer, interior, and exterior buffer layers

- 1) Navigate to the *Analysis Pane* → *Tools*, search for the *Pairwise Buffer (Analysis Tools)* tool.
- 2) Enter the following parameters and click *Run*:

Input Features: XX_WB

Output Feature Class: XX_WB_Buffer and save to your local geodatabase.

Distance [value or field]: Linear Unit, 1, Statute Miles

Method: Planar

Dissolve Type: No Dissolve

Maximum Offset Deviation: 0, Decimal Degrees

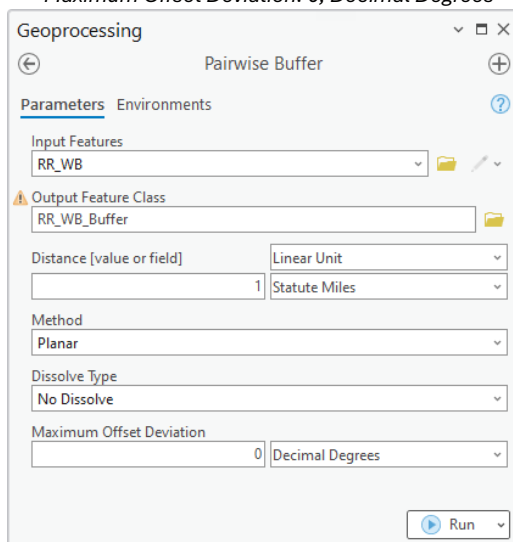


Figure 27. Pairwise Buffer (Analysis Tools) Tool

19 of 38

- 3) Navigate to the *Analysis Pane* → *Tools* and search for *Union*. Click on the *Union (Analysis Tools)* tool.
- 4) Enter these parameters and click *Run*:
 - Input Features*: XX_WB_Buffer (or the buffer layer you created for your watershed), RR_1801_HUC8 (or the polygon shapefile for your watershed)
 - Ranks*: Leave blank
 - Output Feature Class*: XX_WB_Buffer_Union (save to your local geodatabase)
 - Attributes to Join*: All attributes
 - Gaps Allowed*: Check the box

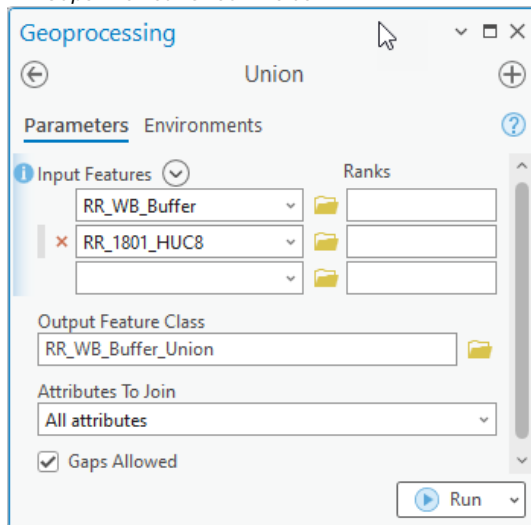


Figure 28. Union (Analysis) Tools

- 5) Right-click on the *XX_WB_Buffer_Union* shapefile in the *Contents* pane and open the attribute table.
- 6) There should be 3 records in the attribute table (highlight each row to identify which is which):
 - a. **Optional:** 1 record that corresponds to the 1-mile sliver surrounding the watershed boundary. Export this record as a separate layer, *XX_WB_Exterior_Buffer* to your local geodatabase.
 - b. **Mandatory:** 1 record that corresponds to the 1-mile sliver within the watershed boundary. Export this record as a separate layer, *XX_WB_Interior_Buffer*, to your local geodatabase.
 - c. **Mandatory:** 1 record that corresponds to the interior polygon within the watershed, 1 mile smaller on every side than the watershed itself. Export this record as a separate shapefile, *XX_WB_Interior*, to your local geodatabase.

19 of 38

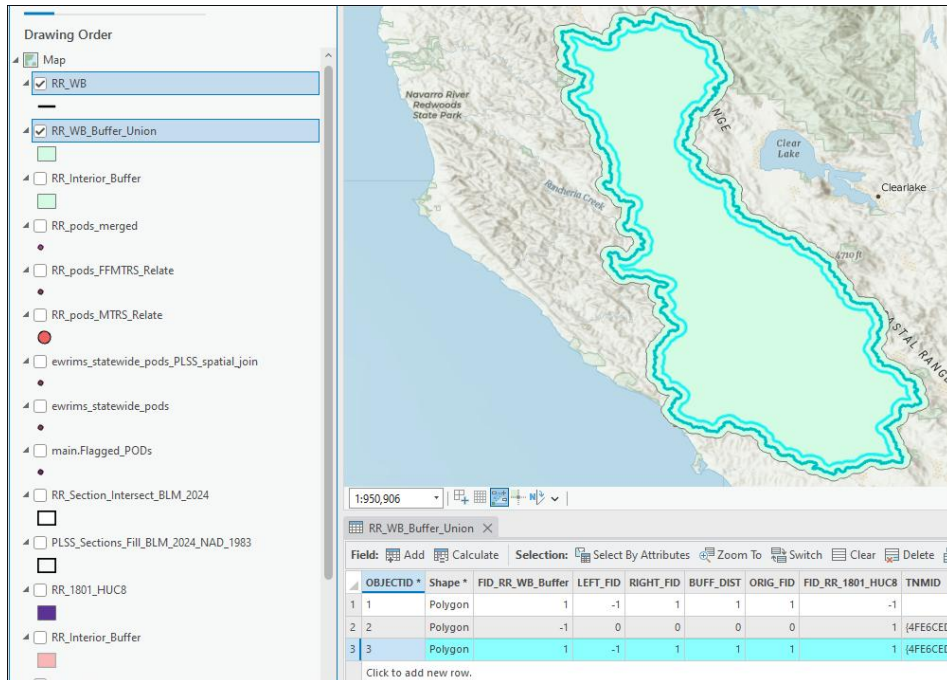


Figure 29. In this image, the “RR_WB_Interior_Buffer” record is selected.

1.5.3 Eliminating Unnecessary PODs from the Manual Review

- 1) Right-click in the Contents pane on the *XX_pods_merged* shapefile and open the attribute table. Add a new field with the name and alias *Case* and data type *Short* into the attribute table.
- 2) Go to the *Analysis* tab → *Tools* → *Select Layer by Location (Data Management Tools)* tool.
- 3) Enter these parameters and click *Run* to execute:
 - Input Features:* *XX_pods_merged*
 - Relationship:* *Intersect*
 - Selecting Features:* *XX_WB_Interior*
 - Search Distance:* Blank and Decimal Degrees
 - Selection Type:* *New Selection*

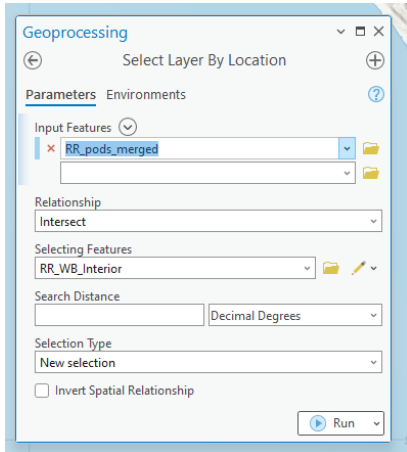


Figure 30. Select Layer by Location Tool

- 4) Once the selection is complete, open the attribute table of the *XX_pods_merged* shapefile and locate the *Case* field.
- 5) Use the field calculator to update the *Case* field value to 0 for the selected points
 - a. Make sure “Use the selected records” is toggled on.
- 6) Run a **definition query** to exclude all Case 0 records from *XX_pods_merged*
 - a. Right click layer > properties > Definition Query > Add new
 - b. Define your query such that Where *Case is null*

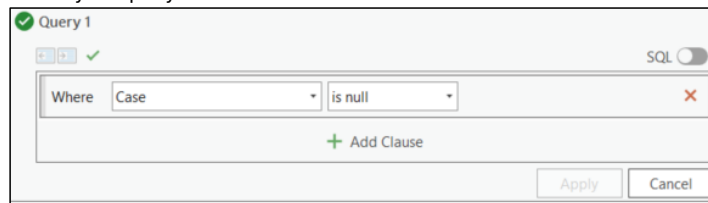


Figure 31. Definition Query for null cases

- 7) Export the selected records as a new shapefile, *XX_pods_manual_review1*.
- 8) Add a definition query to *ewrims_statewide_pods_PLSS_spatial_join* to search for any records where the fields *TRIB_DESC*, *WATERSHED*, or *SOURCE_NAME* contain your watershed name. Here is an example definition query for the Russian—using the *UPPER()* function accounts for both lower-case and upper-case instances.

```
UPPER(TRIB_DESC) LIKE 'RUSSIAN%'
OR UPPER(WATERSHED) LIKE 'RUSSIAN%'
OR UPPER(SOURCE_NAME) LIKE 'RUSSIAN%'
```

- 9) After activating the definition query, export the records in the *ewrims_statewide_pods_PLSS_spatial_join* attribute table to a new shapefile, *XX_pods_manual_review2*, in your local geodatabase.
- 10) Merge *XX_pods_manual_review1* and *XX_pods_manual_review2* into a single shapefile, *XX_pods_manual_review* by using the *Merge* tool.

- 11) Delete the identical records from *XX_pods_manual_review* based on the *POD_ID* field.
- 12) Export *XX_pods_manual_review* as a CSV, *XX_pods_manual_review.csv* to the *DWRAT_DataScraping/Demand/OutputData* folder.

2 Manual Review Process using ArcGIS and Excel

2.1 Import the Manual Review POD List into Excel

- 1) Create an *xlsx* file within your *DWRAT_DataScraping/Demand/OutputData* folder called *XX_pods_manual_review.xlsx*.
- 2) Import *XX_pods_manual_review.csv* into *XX_pods_manual_review.xlsx* (“manual review spreadsheet”) by using Power Query—refer to [Appendix 3.1 Using Power Query and M Scripts](#).
- 3) Navigate to the *RR_GIS_Manual_Review_M_Script.txt* (M Script”) file in the *Scripts* folder of your *DWRAT_DataScraping/Demand* GitHub sub-repository and open it in Notepad++.
- 4) Copy and paste all the content from the *M Script* into the Advanced Editor in Power Query.
- 5) Confirm there are no syntax errors and click *Done*.
- 6) The *M script* makes a bunch of changes to the original CSV that you should see reflected in Power Query:
 - a. LATITUDE gets renamed to *ewrims_latitude*
 - b. LONGITUDE gets renamed to *ewrims_longitude*
 - c. SOURCE_NAM gets renamed to *ewrims_SOURCE_NAME*
 - d. TRIB_DESC gets renamed to *ewrims_TRIB_DESC*
 - e. WATERSHED gets renamed to *ewrims_WATERSHED*
 - f. COUNTY gets renamed to *ewrims_COUNTY*

The *M Script* also adds the following fields:

- g. Report_Latitude
- h. Report_Longitude
- i. Report_Northing
- j. Report_Easting
- k. Report_Zone
- l. Report_Countty
- m. Report_Source Creek
- n. Report_Tributary
- o. Report_Thence
- p. Report_APN
- q. Report_Township
- r. Report_Range
- s. Report_Section
- t. Report_QSection
- u. Report_QQSection
- v. Report_POD_Information
- w. LATITUDE
- x. LONGITUDE
- y. StreamStats
- z. Keep_POD

aa. POD_Justification

- 7) After you apply the *M Script* with Power Query, export the resulting query as the “XX_GIS_Manual_Review” sheet within your spreadsheet. Remember, “XX” is just your 2-character watershed identifier.

2.2 Bulk Download all eWRIMS documents for your watershed

- 1) Create a *Reports* child folder (if it hasn’t already been created) within your *DWRAT_DataScraping/Demand/InputData* folder. The *eWRIMS_Document_Bulk_Downloader.R* script will save all the eWRIMS documents for your watershed to this folder if you follow the remaining steps in this subsection.
- 2) Open *Demand.Rproj* with RStudio.
- 3) Navigate to your local version of the *DWRAT_DataScraping* GitHub repository.
- 4) Open the *Watershed_Demand_Datasets_Paths.xlsx* spreadsheet from the *DWRAT_DataScraping/Demand/InputData* folder.
 - a. Scroll to the *GIS_Preprocessing* section of the spreadsheet and fill out these fields:
 - i. **GIS_PREPROCESSING_SPREADSHEET_PATH**—Enter the full file path of *XX_pods_manual_review.xlsx*.
 - ii. **GIS_PREPROCESSING_WORKSHEET_NAME**—Enter the name of the worksheet containing your manual review records—*XX_GIS_Manual_Review*.
 - iii. **IS_SHAREPOINT_PATH_GIS_PREPROCESSING_SPREADSHEET**: Enter “FALSE” because you saved your spreadsheet locally and not on the SDA SharePoint.
 - iv. **EWRIMS_REPORTS_FOLDER_PATH**—Enter the full file path for your *Reports* folder.
 - b. Save and close the spreadsheet.
- 5) Navigate to the *eWRIMS_Document_Bulk_Downloader.R* script and run the entire script. The script is designed to skip any reports that have already been downloaded for your watershed. Consequently, you won’t waste time if you have to pause the script halfway—when you resume, the script will just pick up where you left off. It is unlikely that the number of PDFs in your *Reports* folder will perfectly match the number of PODs in *XX_pods_merged_YYYY-MM-DD.csv* (your full POD list) because:
 - a. Some water rights have multiple PODs and there is only 1 eWRIMS PDF document for each water right.
 - b. A few water rights only have a paper record that has not been scanned into eWRIMS yet.
 - c. A few water rights have corrupt PDFs.

Commented [PA3]: This script is failing because the APPLICATION_NUMBER contains HUC12 values instead of water right IDs—go back into the ArcGIS Pro project and figure out why this is happening.

2.3 How to fill out the manual review spreadsheet

- 1) Now that you have downloaded the eWRIMS PDFs, you can begin the manual review by referring to the PDF for each water right to fill out the various *report fields*. The figures below demonstrate where to look for each *report field*.

STATE OF CALIFORNIA
 CALIFORNIA ENVIRONMENTAL PROTECTION AGENCY
 STATE WATER RESOURCES CONTROL BOARD

DIVISION OF WATER RIGHTS

PERMIT FOR DIVERSION AND USE OF WATER

PERMIT 20828

Application 30245 of Carolyn L. Pride

4026 Spring Mountain Road, St. Helena, CA 94574

filed on April 19, 1993, has been approved by the State Water Resources Control Board
 SUBJECT TO PRIOR RIGHTS and to the limitations and conditions of this permit.

Permittee is hereby authorized to divert and use water as follows:

1. Source:

Tributary to: Report_Tributary

Unnamed Stream

Ritchie Creek thence

Napa River thence

San Pablo Bay Report_Thence

Report_Section Report_Township Report_Range Report_Meridian

2. Location of point of diversion <u>Report_POD_Information</u>	40-acre subdivision of public land survey or projection thereof	Section	Township	Range	Base and Meridian
North 1,000 feet and West 1,500 feet from SE corner of Section 19	SW $\frac{1}{4}$ of SE $\frac{1}{4}$	19	8N	6W	MD
<u>Report_QQ Section</u>	<u>Report_Q Section</u>				

County of Napa

SWRCB 14 (6-94)

Report_County

Figure 32. Locating the Report Fields on an eWRIMS document (Part 1)

Initial Statement of Water Diversion and Use
Page 2 of 4

Section 1: Type of Claim
(Select all that apply to the type of claim(s) under which you are diverting water)
☒ Riparian ☐ Pre-1914 ☐ Court Decree ☐ Pending Appropriative Application ☐ Pueblo ☐ Other: _____
*If you checked Court Decree, Pending Appropriative Application, or Other, list the Decree Number, Application ID Number or Status or provide an explanation
List any related existing water rights, if applicable (e.g. Appropriative Water Right ID: A012345)

Section 2: Water Course Description
Water Course Name at the Point of Diversion (POD): Mill Creek
Water Course is tributary to: Russian River
This report lacks a Report_Thence field, so leave it blank in the spreadsheet

Section 3: Point of Diversion and Legal Land Description
Provide the location of the POD using one of the following methods (check one box and enter data if applicable)
☒ Latitude/Longitude Measurements: Latitude: 39.119400 Longitude: -123.119660
☐ California Coordinate System (NAD1983) North: Report_Northing East: Report_Easting Zone: Report_Zone
☐ USGS Topographic Map with Point of Diversion labeled on map (if checked, map must identify point of diversion)
Assessor's Parcel Number (APN) where Point of Diversion is located (if APN has been assigned): 189-070-24-00 County: Mendocino
Provide Public Land Description to nearest 40 acres (if assigned)
SW Report_QQ_Section Report_QSection Report_Section NE Report_Section Report_Township Report_Range Township 15N Range 11W B&M Mount Diablo
Report_Meridian

Section 4: Map Description
(Check boxes indicating each map to be provided)
Identify the location of the place of use on a specific United States Geological Survey (USGS) Topographic Map, or County Assessor's parcel map or any other maps with identifiable landmarks. If assigned, provide the public land description to the nearest 40-acre subdivision and the assessor's parcel number.
☐ USGS Topographic Map ☐ County Assessor's Parcel Map ☒ Map with identifiable landmarks
Provide a general description of the area in which the water was used (e.g. Domestic water supply for house, and irrigated crops, campground, etc.)
Domestic water supply
Assessor's Parcel Number(s), where the water was used (if APNs have been assigned)
189-070-24-00

Section 5: Purpose of Use Description
(Select all that apply)
☐ Irrigation Number of acres: _____
☒ Domestic Maximum number of persons served: 2
☐ Stock watering Number and type of stock: _____
☐ Other Explain: _____

S027304

Figure 33. Locating the Report Fields on an eWRIMS document (Part 2)

- 2) Note that not all eWRIMS PDFs have the same design—the water rights form has evolved over the years and different types of rights have different fields. Additionally, even after you apply optical character recognition (OCR) to these PDFs, you still may be

unable to apply Ctrl+ F to search for terms because of poor scan quality, especially for older water rights forms. Furthermore, you also do not need to fill out every single field that you added to your manual review spreadsheet with the *M script* (hosted in the *DWRAT_DataScraping/Demand/Scripts* folder). If the report already explicitly states the latitude and longitude of the PODs, then you do NOT need to note all the other *supplemental* geolocational information. The purpose of this supplemental information is to derive the POD latitude and longitude if absent from the report.

3) Filling out the Fields in the Manual Review Spreadsheet

- a. *Report_Latitude* and *Report_Longitude*: If the eWRIMS report has the latitude and longitude, then fill out these two fields. If the coordinates are not provided in decimal degrees, then convert them to decimal degrees before entering them into the spreadsheet. The purpose of these fields is to gauge the accuracy of the eWRIMS GIS coordinates (*eWRIMS_Latitude* and *eWRIMS_Longitude*)—if *eWRIMS_Latitude* and *eWRIMS_Longitude* are accurate, then set the *LATITUDE* and *LONGITUDE* fields in the manual review spreadsheet equal to *eWRIMS_Latitude* and *eWRIMS_Longitude*. If they are inaccurate, then instead populate the *LATITUDE* and *LONGITUDE* fields with the *Report_Latitude* and *Report_Longitude*.
- b. *Report_Northing* and *Report_Easting*: Northings and eastings are coordinates in the state plane coordinate system. The California version of the state plan coordinate system is known as the *California Coordinate System*. If the eWRIMS report lacks the *Report_Latitude* and *Report_Longitude* fields, then look for the *Report_Northing* and *Report_Easting*. Check that the diverter reported the northing and easting in units of feet and note the *Report_Zone* as well. The *Report_Zone* is always established with respect to a coordinate reference system (CRS), although this CRS may not be explicitly stated.

Explicit CRS example

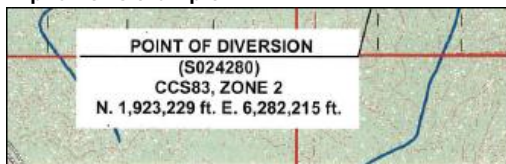


Figure 34. CCS83, ZONE 2 indicates that the CRS is NAD 1983—the CCS is short for California Coordinate System.

Report_Zone with no CRS

Occasionally reports do not indicate the CRS (Payman Alemi was unable to find an example after a cursory search but has seen it happen before). However, the CRS can usually be determined based on context. Northing and easting can be provided with either NAD 1927 or NAD 1983 CRS, but if you convert both sets of coordinates into WGS 84 and plot them in ArcGIS Pro against your watershed polygon shapefile, you can determine which CRS makes more sense. If no *Report_Zone* is listed in the report, then you will need to rely on other geolocational parameters to back-calculate the *Report_Latitude* and *Report_Longitude*. For detailed instructions on this back-calculation process, refer to Appendix [3.3](#).

- c. *Report_Source_Creek*, *Report_Tributary*, and *Report_Thence* fields: The purpose of filling out these fields is to assess whether the POD flows into a stream that belongs to the watershed of interest. If none of these fields have “Russian River” for example, then it is likely that the POD flows into another watershed. As an extra check, enter the *Report_Latitude* and *Report_Longitude* into the USGS StreamStats tool.
- d. *Report_Meridian*, *Report_APN*, *Report_Township*, *Report_Range*, *Report_Section*, *Report_QSection*, *Report_QQSection*, and *Report_POD_Information* are only necessary to populate when the report lacks any type of POD coordinates, or you suspect that the coordinates are incorrect. Additionally, not every report may have all these fields, but even with a handful of them, you can roughly pinpoint the POD coordinates by using ArcGIS Pro. Appendix 3.4 provides a detailed example of this process by using the POD for water right A028127.

In a nutshell, the steps above safeguard against the possibility that the coordinates in eWRIMS were radically misplotted—our ArcGIS Pro analysis has already determined that the eWRIMS GIS coordinates are located within the watershed. However, if the report coordinates differ substantially from the eWRIMS GIS coordinates, we must check that they are still located within the watershed. If they are NOT located within the watershed, then type “No” in the *Keep_POD* field in the manual review spreadsheet because we want to exclude these PODs from the final demand dataset.

- e. *StreamStats* field—Fill out this field with “In watershed” or “Outside watershed” depending on the results of the USGS StreamStats Tool (see Appendix 3.2). The purpose of this field is to catch PODs that are geospatially within the watershed but that flow into another watershed—this is rare but possible. All “outside watershed” records should have their corresponding *Keep_POD* field set to “No” because they should also be excluded from the final demand dataset.
 - f. *Keep_POD*: Indicate as “Yes” or “No”—you either keep the POD or exclude it from the final POD list for the watershed.
 - g. *POD_Justification*: Justify your decision to keep or exclude the POD and be as detailed as possible (cite coordinate system discrepancies and other issues you came across for example)
 - h. *Staff*: Type your name.
 - i. *Review_Date*: Type the date you conducted the manual review for the POD.
- 4) Keep in mind that the manual review spreadsheet that you created from ArcGIS Pro contains only a subset of the PODs in the watershed—this is the subset that might possibly have geolocational errors. Consequently, you need to remove the errant PODs you identified in the previous steps from the full POD list in the *XX_pods_merged_YYYY-MM-DD* csv.
- a. Using Power Query, import the *XX_pods_merged_YYYY-MM-DD* csv into your *manual review spreadsheet*.
 - b. Remove all PODs that were identified for removal during the manual review process and save this filtered list as a separate sheet called *Final_List*.

- 5) Type the full name of the *[ID]_GIS_Preprocessing.xlsx* spreadsheet and the *Final_List* in *Watershed_Demand_Datasets_Path.xlsx*:
 - a. POD_APPLICATION_NUMBER_SPREADSHEET_PATH: this is the file path of your manual review spreadsheet.
 - b. POD_APPLICATION_NUMBER_WORKSHEET_NAME: set to “Final_List”
 - c. IS_SHAREPOINT_PATH_POD_APPLICATION_NUMBER_SPREADSHEET: set to “FALSE”.
- 6) You can now jump to **Section 7. Preliminary Demand Dataset** of the main [Demand Dataset Procedure](#) document.

3 Appendix

3.1 Using Power Query and M Scripts

1. Navigate to *Data* → *Get Data* → *Launch Power Query Editor*

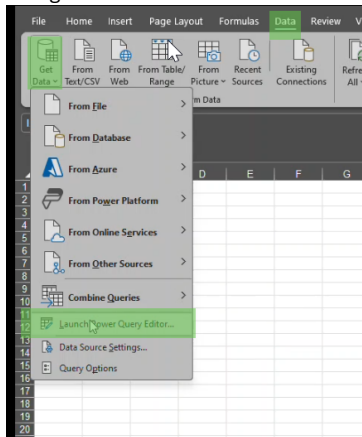


Figure 35. Launch Power Query Editor

2. Navigate to the *Home* Tab → *New Source* → *File* → *Text/CSV* and load the manual review spreadsheet that you generated in ArcGIS Pro [*XX_pods_manual_review.csv*] by clicking *Import* and then *OK* on the resulting dialog boxes.

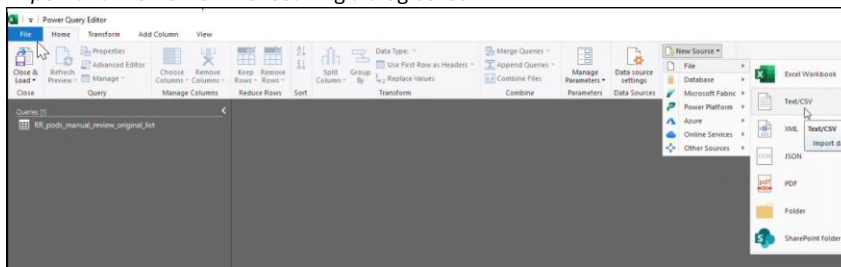


Figure 36. Importing CSVs into Power Query

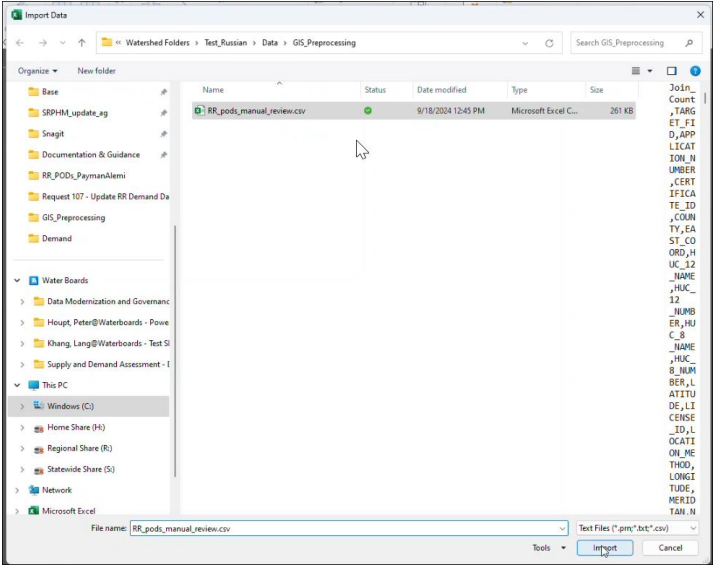


Figure 37. Import Data Window

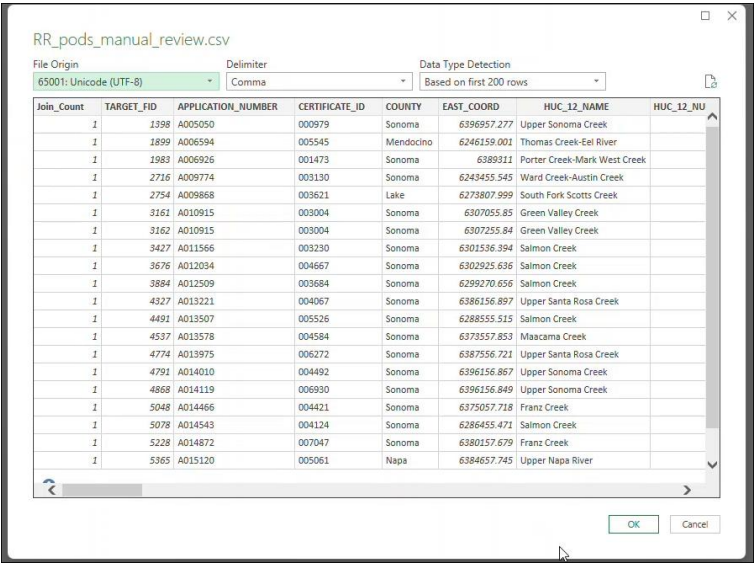


Figure 38. Loading RR_pods_manual_review.csv into Power Query Editor

3. Rename the CSV ("query") that you just imported to *XX_pods_manual_review_original* ("RR_pods_manual_review_original" in this example) by right-clicking on it.

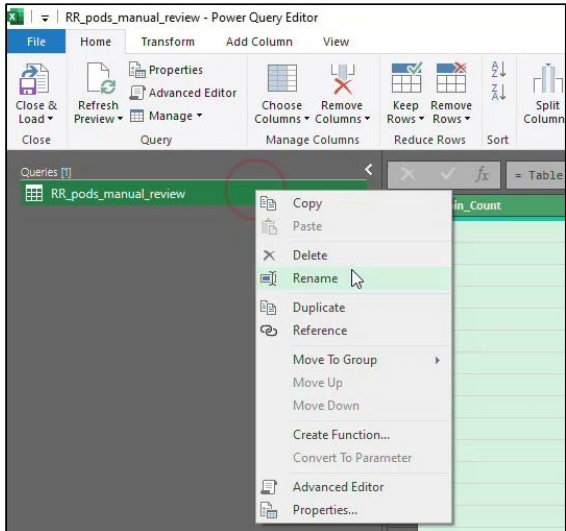


Figure 39. Renaming the query

4. Right-click on the query again and select *Advanced Editor*.
5. Delete everything from the resulting *Advanced Editor* window:

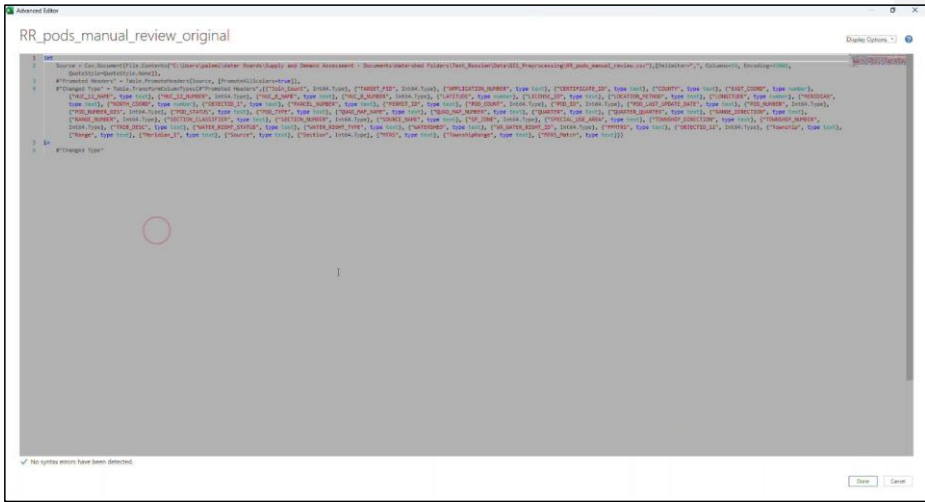


Figure 40. Advanced Editor in Power Query Editor

6. Load the M script of interest ("RR_GIS_Manual_Review_M_Script.txt" in this example) in Notepad++.
7. Copy and paste all the content from the M Script into the *Advanced Editor* window.
8. Adjust the portion of the M Script that defines the local user folder to point to yours:
let

31 of 38

Source = Csv.Document(File.Contents("C:\Users\palemi\Water Boards\Supply and Demand Assessment - Documents\Watershed Folders\Test_Russian\Data\GIS_Preprocessing\RR_pods_manual_review.csv"))

9. Confirm there are no syntax errors and click *Done*.
10. Load the query as a sheet with the same name into Excel by navigating to *Home* → *Close and Load* in the ribbon.

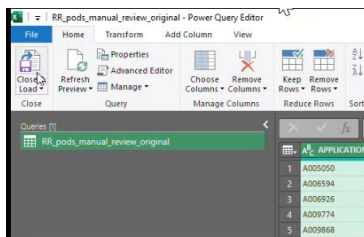


Figure 41. Close and Load the query

3.2 Using the USGS StreamStats Tool

- 1) Go to <https://streamstats.usgs.gov/ss/> and enter the latitude and longitude of the POD of interest into the search box:

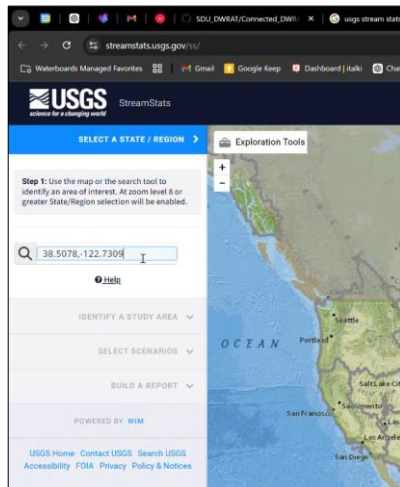


Figure 42. Entering Coordinates in StreamStats

- 2) Then select the automatic coordinates that appear in the dropdown menu:

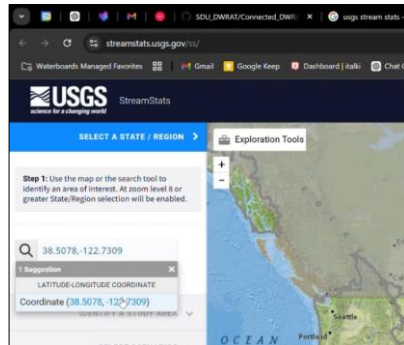


Figure 43. Coordinate Dropdown option

- 3) Next click *Exploration Tools* and scroll down to *Flow (Raindrop) Path*:

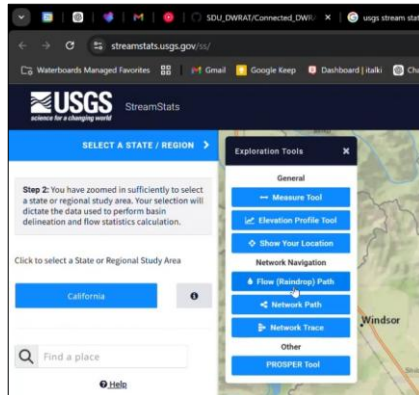


Figure 44. Flow (Raindrop) Path

- 4) After that, click *Select Point on Map*, pan to the map and then select the red point.

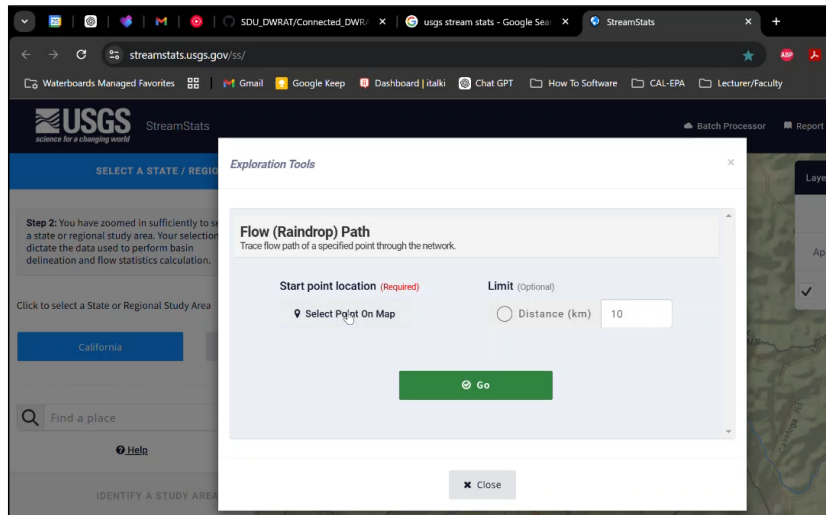


Figure 45. Flow (Raindrop) Path



Figure 46. Selecting the point on the map

5) The Flow (Raindrop) Path pop-up menu will re-appear and now click Go.

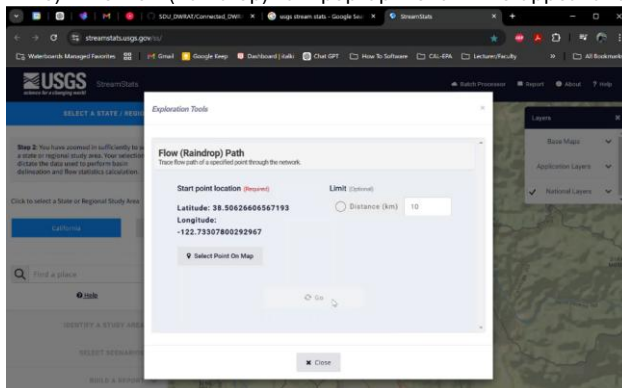


Figure 47. Flow (Raindrop) Path—Click Go

6) Zoom to the end of the red flow path, read the map carefully and confirm that the final watercourse name matches the name of the watershed. In this example, the

34 of 38

final watercourse name is the *Russian River*, which confirms that the POD flows into the *Russian River* watershed:



Figure 48. Watershed Outlet in StreamStats

3.3 Using EarthPoint to convert Northings/Easting into Latitude/Longitude

- 1) Browse to <https://www.earthpoint.us/StatePlane.aspx> and enter these parameters and click *Calc.*
XY Unit Measure: Us Survey Feet
Zone Number: 0402 or select 0402 – California Zone 2 from the dropdown list.
Easting/X: 6282215
Northing/Y: 1923229
- 2) Scroll down the page to find the latitude and longitude in decimal degrees:

Calculated Values - based on Degrees Lat Long to seven decimal places.	
Position Type	State Plane - California Zone 2
Degrees Lat Long	38.4390900°, -122.9756534°
Degrees Minutes	38°26.34540', -122°58.53921'
Degrees Minutes Seconds	38°26'20.7239", -122°58'32.3524"
State Plane X Y (Meters)	0402 1915822.962mE 586201.372mN
X Y (US Survey Feet)	0402 6282215.000UsFtE 1923229.000UsFtN
X Y (International Feet)	0402 6282227.564ftE 1923232.846ftN
UTM	10S 502124mE 4254535mN
UTM centimeter	10S 502124.75mE 4254535.07mN
MGRS	10SEH0212454535
Grid North	0.0°
GARS	115LS11
Maidenhead	CM88MK25WJ11
GEOREF	DJNJ01462634
Plus Code	84CVC2QF+JP
Plus Code Extended	84CVC2QF+JPQCVVW
what3words	relief.bargained.chatting

Figure 49. Northing/Easting Conversion to Latitude and Longitude with Earthpoint

- 3) In ArcGIS Pro, confirm if your coordinates make sense and match the existing eWRIMS coordinates—navigate to the *Map* tab and click on the *Go to XY* entry tool.

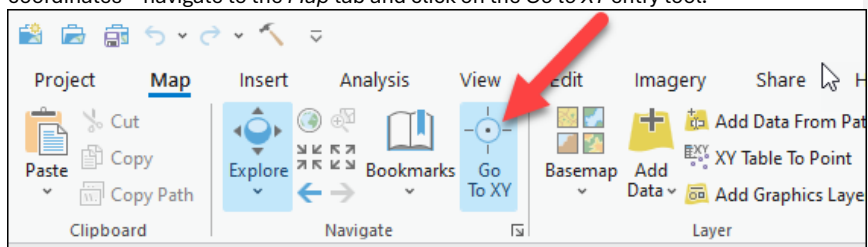


Figure 50. Go to XY Tool

- 4) Add an XY label to the map and flash the point. Use the pan and zoom features to bring the point into view. Select the [WS]_pods_manual_review shapefile and look for the nearest PODs. In this case, as [Figure X] demonstrates, the report coordinates match identically with the eWRIMS coordinates for the water right S024280.

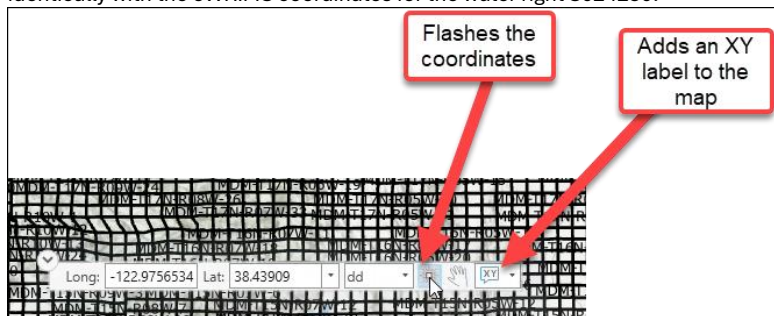


Figure 51. Features of XY Tool

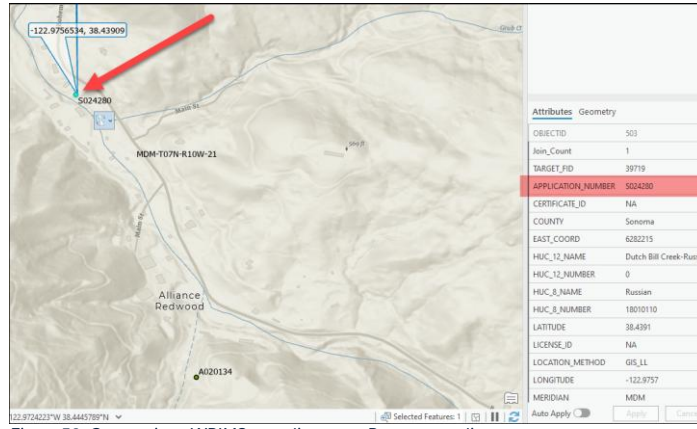


Figure 52. Comparing eWRIMS coordinates to Report coordinates

- 5) Type the latitude and longitude that you derived from the EarthPoint website into the *Report_Latitude* and *Report_Longitude* fields but indicate with comments (right-click on each cell to create a comment) that these values were not listed in the report but were derived from the Northing and Easting.

3.4 Finding POD Coordinates in the absence of latitude/longitude and northing/easting

- 1) Taking water right A028127 as an example, we open the report and look for the POD location description.

THE POINT OF DIVERSION OF SUCH WATER IS LOCATED:

North 1,000 feet and West 600 feet from SE corner of Section 17, T17N, R13W, MDBM, being within SE¼ of SE¼ of said Section 17.

Figure 53. A028127 POD location description

In this example, no Report_APN is listed but we have sufficient information to pinpoint the POD in eWRIMS because we have values for these fields:

- Report_Meridian: MDBM (“Mount Diablo Base and Meridian”)
- Report_Township: T17N (or “17N”)
- Report_Range: R13W (or “13W”)
- Report_Section: 17
- Report_QSection: SE (“Southeast”)
- Report_QQSection: SE
- Report_POD_Information: North 1000 ft and West 600 ft from SE corner of Section 17, 17N, 13W MDBM—you can use these measurements to pinpoint the exact location in ArcGIS Pro, but you don’t have too—eyeballing the QQSection is sufficient.

- 2) Make sure that you have loaded these shapefiles into ArcGIS Pro:
 - *RR_QQ_Sections* (this is derived from a larger statewide QQ_Section shapefiles but I whittled it down to the Russian River for this example)

37 of 38

- *PLSS_Sections_Fill_BLM_2024_NAD_1983* (this is the PLSS layer for the entire state of California)
- *RR_pods_manual_review*

3) You can expedite the process by assuming that the coordinates plotted in eWRIMS are accurate (even though the report itself lacks coordinates); the vast majority of the time, eWRIMS coordinates are correct. This allows us to open the attribute table of the *RR_pods_manual_review* shapefile, select the record for A028127 and zoom to the selected record:

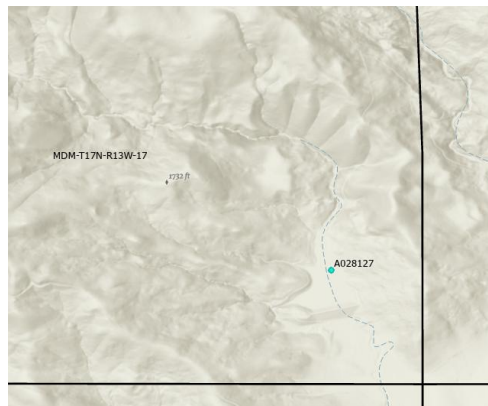


Figure 54. A028127 POD superimposed on statewide PLSS layer

Figure 54 confirms that the POD is indeed located in Section 17 of Township 17N, Range 13W. If we zoom in, we can visually confirm, as shown in Figure 55, that this POD is located within the SE quarter of Section 17.

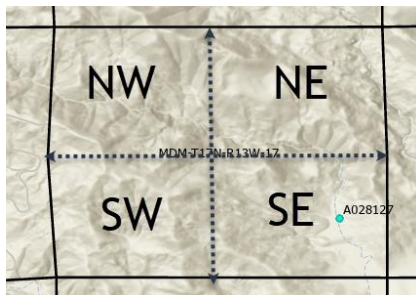


Figure 55. Dividing Section 17 into Quarter-Sections

After we load the *QQ_Sections* shapefile, and zoom in further, as shown in Figure X, we visually confirm that the POD is located within SE quarter of the SE quarter-section.

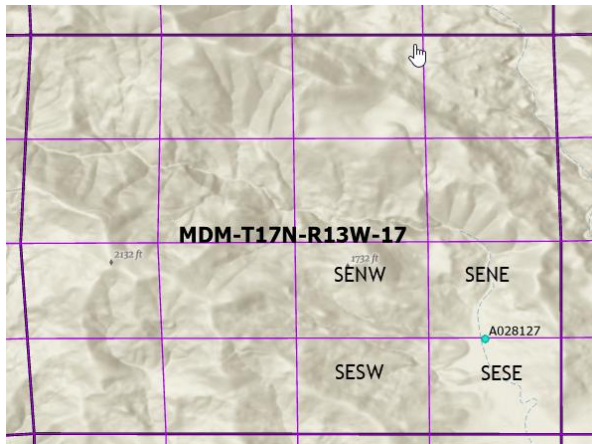


Figure 56. A028127 superimposed on quarter-quarter sections

For our purposes, this is accurate enough and it validates our initial assumption that the eWRIMS coordinates are accurate. Consequently, we can set the final LATITUDE and LONGITUDE fields in the manual review spreadsheet equal to the eWRIMS_Latitude and eWRIMS_Longitude fields.